

Lab 2

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Chosen system: Italian Restaurant

RE activity: Specify Requirement

Complete prompt:

“

I would like you to write your response to the following task in the attached markdown file.

RE Activity: Specify Requirement

****AI's role**:** Requirements specification expert

****My role**:** Requirements engineer

****My purpose**:** Requirement formulated in an optimized way to increase readability and comprehensibility.

****Task**:** Optimize the formulation of the following requirement.

Formulation standard: ****User Story****

Quality standard: ****INVEST CRITERIA****

Step 1: If the formulation standard is "none," identify the standard used for the input requirement (e.g., use case, user story incl. acceptance criteria, IEEE, SOPHIST template). If a standard is found, use it for further proceedings.

Step 2a: If the formulation standard is still “none”, select the most appropriate standard, such as use case or user story, and use it.

Step 2b: Otherwise, if a formulation standard has been identified, determine if there are more appropriate formulation standards for this requirement. If so, stop proceeding and suggest one or two alternatives. Then, ask which formulation standard should be used. Proceed once the formulation standard has been given. The provided standard may be the one originally entered.

Step 3: Formulate the requirement according to the chosen standard.

Step 4: Validate the requirement according to the quality standard. If the requirement does not meet any quality criteria, request missing information and repeat step 3.

****Format**:** *Plain Text*

****Input**:** ****Existing:** customers already order food & drinks

online and pick up in-store about 30 minutes later.

New: at checkout, customers can choose pickup
or delivery.

Delivery payment: paid online via credit card, direct bank
transfer, or PayPal; customer provides a delivery address.

Delivery operations: 10 km radius, 2–3 riders on e-bikes,
guaranteed delivery within 70 minutes of order placement.

Routing & app: the system auto-generates routes (item list + turn-by-turn directions via
Google Maps);

riders use a smartphone app and are assigned a route as soon as they mark themselves
available.

Side note: the owner's son is considering opening his own restaurant in 1–2 years and
would like to reuse

this system.**

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Outputs from part 3:

Requirements Specification

Step 1: Identify the formulation standard

The input is informal business and system information. It is not already written as a complete user story, use case, or formal specification. The requested formulation standard is User Story.

Step 2: Check whether another standard is more appropriate

The User Story standard is appropriate for the customer-visible capabilities because it expresses who needs a capability and why. However, the input also contains operational constraints and technical behaviour, such as delivery radius, delivery time, route generation, payment methods, and future reuse. These details are better represented as separate acceptance criteria and supporting system requirements rather than one large user story.

Because the requested standard is User Story and remains suitable, the requirement is decomposed into several related user stories. No change of standard is required.

Step 3: Optimized user stories

US-1: Select a fulfilment method

As a customer,

I want to choose store pickup or delivery at checkout,
so that I can receive my order in the way that suits me.

Acceptance criteria:

- At checkout, the customer can select exactly one fulfilment method: pickup or delivery.
- For pickup, the existing in-store pickup process remains available.
- For delivery, the system requests and validates a delivery address before the order is placed.
- The selected fulfilment method is shown in the order confirmation.

US-2: Pay for a delivery order

As a customer ordering delivery,

I want to pay online by credit card, direct bank transfer, or PayPal,
so that I can complete my order without paying the rider.

Acceptance criteria:

- The customer can select credit card, direct bank transfer, or PayPal as the online payment method.
- The order is not confirmed as paid until the selected payment method reports success.
- A failed or cancelled payment does not create a confirmed delivery order.
- The customer receives confirmation of the selected payment method and payment status.

US-3: Check delivery availability

As a customer,

I want the system to tell me whether delivery is available for my address,
so that I know whether I can place a delivery order.

Acceptance criteria:

- The system accepts a delivery address during checkout.
- The system allows delivery only when the address is within the restaurant's defined 10-kilometre delivery radius.
- The system clearly informs the customer when the address is outside the delivery area.
- The system does not accept an outside-area address as a delivery order.

US-4: Receive a delivery-time commitment

As a customer ordering delivery,
I want to receive my order within 70 minutes of placing it,
so that I can plan when to expect my food and drinks.

Acceptance criteria:

- A delivery order records its placement time.
- The system calculates and displays a promised delivery time no later than 70 minutes after order placement.
- The customer can see the delivery status and promised time after placing the order.
- An order that cannot meet the 70-minute commitment is flagged for operational handling and the customer is informed.

US-5: Receive an automatically generated delivery route

As a delivery rider,
I want to receive an assigned route containing my delivery item list and turn-by-turn directions,
so that I can deliver each order efficiently.

Acceptance criteria:

- The system generates a route for an eligible delivery order.
- The assigned route includes the relevant delivery item list.
- The assigned route includes turn-by-turn directions provided through Google Maps.
- The route is available to the rider in the smartphone application.

US-6: Receive a route when becoming available

As a delivery rider,

I want to mark myself as available in the smartphone application,
so that the system can assign me a delivery route.

Acceptance criteria:

- A rider can mark themselves available in the smartphone application.
- The system assigns an eligible route to an available rider as soon as the rider becomes available.
- The system assigns routes only to riders who are available and operating within the configured delivery operation.
- The rider can view the assigned route and its item list in the application.

US-7: Configure the delivery service for reuse

As a restaurant operator,

I want the delivery service's restaurant-specific settings to be configurable,
so that the system can be reused by another restaurant without changing its core software.

Acceptance criteria:

- The delivery radius, delivery-time commitment, payment methods, fulfilment methods, and rider settings can be configured per restaurant.

- Restaurant-specific orders and delivery routes are kept separate from those of other restaurants.
- A new restaurant can configure its service without modifying the core delivery functionality.

Operational constraint:

The delivery operation must support 2 to 3 riders using e-bikes. This is a constraint on the operation and should not be treated as a user story unless a specific rider or operator capability is required.

Step 4: INVEST validation

The stories are valuable because each describes a clear customer, rider, or operator outcome. They are negotiable because the implementation details, such as the exact route-assignment algorithm, are not prescribed beyond the stated business needs. They are estimable and small because the original compound requirement has been divided into distinct capabilities. They are testable because each story has observable acceptance criteria.

The stories are sufficiently independent for analysis and prioritisation, although the delivery stories have implementation dependencies: delivery availability precedes delivery payment and fulfilment, and route assignment depends on a confirmed delivery order. These dependencies do not prevent the stories from being independently understood or tested.

Clarifications required before final baselining:

1. Is the 10-kilometre radius measured from one restaurant, from each restaurant, or from a configurable delivery location?
2. Does the 70-minute guarantee begin when the customer submits the order, when payment succeeds, or when the restaurant accepts the order?

3. What should happen when no rider is available or when all 2 to 3 e-bike riders are occupied?
4. Which payment provider and bank-transfer flow should be used, and what counts as successful payment for a bank transfer?
5. Is Google Maps a mandatory integration, and which Google Maps product and account will provide the directions?
6. For future reuse, should the system support multiple restaurants at the same time, or only be configurable for one restaurant deployment at a time?

Until these points are answered, the stories satisfy the INVEST criteria at a functional level, but the delivery-time, routing, payment, and reuse requirements cannot be considered fully unambiguous and ready for implementation.

My reflective discussion (part 4):

This assignment really put into perspective just how quick AI can be for this sort of planning – it fired out 7 user stories in no time, which to be honest is more than I would have thought of even if I had all day.

Looking at the results, the AI did a really good job and stuck tightly to the prompt instructions. It took my template paragraph and neatly turned it into 7 different user stories using the standard structure ("As a... I want... so that..."). It correctly picked out all of the different personas for each story – customer, e-bike rider, restaurant owner etc. Each story came with concrete acceptance criteria using INVEST principles, making them small, testable and easy to use for estimating in sprints.

In terms of completeness, it managed to capture all of the features that were mentioned in the original notes, such as the 10km range check, 70-minute guarantee, online payment options, Google maps routing etc. However, there are still a few gaps I noticed if you're thinking about applying this to a real world system. The AI didn't account for practical edge cases, such as what happens if all three riders are already busy when a new order comes in, or what happens if a rider loses access to their phone while en-route. It also left out key non-functional requirements like GDPR privacy for storing customer info like address and location data, and payment processing.

Overall, I found this process extremely useful for planning real RE work. Generating a detailed draft backlog in seconds saves a huge amount of time because I have witnessed how painful this can be while I was on placement. For me, the most valuable

part was actually the list of six clarification questions the AI produced at the end. In a real-life project, I could take those exact questions straight into a meeting with the restaurant owner to figure out exactly what they need, because these are the sort of things I would struggle to think of myself. This task really showed me that while AI is great at speeding up the heavy lifting and initial drafting, you still need a human at the end of the day to spot stuff like edge cases, and to have the ability to personally go and check with actual stakeholders.